The Synapse - Signal Transmission

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

At a chemical synapse, an incoming action potential is not transmitted directly to the next cell; first it is converted into a chemical signal. When the action potential reaches the terminal button, voltage-gated Ca2+ channels open in the presynaptic membrane. Ca2+ ions flow into the terminal button and trigger vesicles to move to and fuse with the presynaptic membrane. Acetylcholine is released into the synaptic cleft (= exocytosis). The acetylcholine diffuses to the postsynaptic membrane and binds there to ligand-gated Na+ channels. These open, allowing Na+ ions to flow into the downstream cell. This changes the membrane potential of the postsynaptic membrane. If the change is large enough, excitation can be triggered again in the following cell. To prevent the transmitter's effect from persisting, acetylcholine is enzymatically broken down in the synaptic cleft by acetylcholinesterase. The components are then taken up again into the terminal button, regenerated and reused there. In this way, chemical signal transmission at the synapse is precisely limited in space and time.

Task 1: Label the structures of the synapse

Complete the labelling task in the "Synapse" mode. Export the correctly labelled figure and insert it below.

Insert here

Task 2: Sequence of signal transmission

Put the sequence of processes at the synapse in the correct order. Then export the solved task and insert it below.

Insert here

Task 3: Explanation of signal transmission

Describe how a signal is transmitted from the terminal button through the synaptic cleft to the next cell.